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SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Duration: 60 Minutes

Total Marks: 25

Code of Conduct C1925211277

I S. Rowith (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Criteria	Conceptual Accuracy (1.5)	Coverage of Concepts (1)	Use of Examples (0.75)	Comparison / Differentiation (1)	Clarity of Expression (0.75)	TOTAL (5)
Weightage	30%	20%	15%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

Assessment Tasks

Short Answer Test

1. Describe the evolution from standalone computing to cloud-based systems. Summarize key technological transitions that enabled cloud computing.
2. Interpret the working of hypervisors in virtualization. Differentiate between Type 1 and Type 2 hypervisors with examples.
3. Describe the role of APIs in cloud service integration. Explain how APIs support interoperability between services.
4. Explain the concept of resource pooling in cloud computing. Illustrate how it benefits multiple users.
5. Interpret the concept of rapid elasticity in cloud computing. Explain how it supports dynamic workloads.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

1. Standalone computing

Initially computers worked independently. Data and applications were stored in a single machine, making sharing and remote access difficult.

Client server computing.

Applications and data were stored on central servers, allowing multiple clients to access resources over a network.

Distributed computer.

Several computers worked together to complete tasks, improving performance and reliability.

Grid computing:

Resources from different locations were combined to solve large computing problems.

cloud computing.

Computing resources such as servers, storage, database and software are provided over the Internet on .

Key technological transitions .

- * High speed Internet .
- * Virtualization technology .
- * Data Centres .
- * Web services and APIs
- * Broadband networking

Example :

Google drive allows users to store and access files from any device through the cloud .

2 .

A hypervisor is software that create and manages virtual machines (VMs) . It allows multiple operating system to run on one physical computing by sharing hardware resources .

TYPE 1 Hypervisor	TYPE 2 Hypervisor
Runs directly on hardware	Runs on top of a host operating system
Faster and more secure	slightly slower
used in data centres	used for testing and personal use
Ex : VMware ESXi , microsoft hyper v	Ex : Oracle virtual Box , VMware workstation

Example :

A company use VMware ESXi to run many virtual servers on one physical

An API (Application Programming Interface) is a set of rules that allows different software applications to communicate.

Role in APIs in cloud computing.

- * Connect different cloud services.
- * Enables application integration.
- * Automate cloud operations.
- * Allows developers to access cloud resources.

How API supports interoperability.

- * Exchange data between different platform.
- * Allows services from different cloud providers to work together.
- * Improve flexibility and scalability.

Example.

- * A weather application uses a cloud API to display live weather information.

4. Resource pooling is a cloud computing feature where computing resources (servers, storage, memory and network) are shared among multiple users -

Benefits .

- * Efficient resource utilization .
- * Reduced cost .
- * High availability .
- * Better scalability .
- * Supports multiple users simultaneously .

Example .

Amazon web services (AWS) shares its servers among many customers while keeping their data separate and secure .

5. Rapid Elasticity is the ability of cloud computing to automatically increase or decrease computing resources based on workload demand .

How it supports dynamic workloads .

- * Increases resources during high traffic .
- * Reduces resources when demand decreases .
- * Saves cost by using only required resources .
- * Maintains application performance .